BirdShift

Context-Debiased Fine-Grained Bird Recognition under Geography, Background, and Long-Tail Shift

Progress report: experiments and results

DSE-312 Computer Vision
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At a glance

What has been built and measured since the midterm

84.55%
CUB test top-1, ResNet-50 at 448 px.
Matches the published baseline (84.5%)

−0.23 pp
BirdShift drop when the background is replaced
(full image: −1.11 pp)

23 → 50%
Tail-class accuracy with logit-adjusted loss
(+9.3 pp overall)

−19.3 pp
Unseen region: 81.9 → 62.7%
(unseen background: −1.7 pp)

- Three input variants trained and compared: **full image**, **bird crop**, **BirdShift** (background suppression during training).
- Controlled background-shift test set: original, blurred and replaced backgrounds on the same birds.
- Long-tailed CUB split (10:1) with four imbalance methods, reported by head / medium / tail.
- Training recipe fixed: torchvision weights + SGD + cosine LR gives +4.2 pp test over our first recipe.
- Two real held-out splits: an unseen **region** (iNaturalist) and an unseen **background type** (CUB clusters).

Data and splits

One fixed protocol for every run

Split	How it is built	Images
CUB train	85% of the official train split, stratified by species, seed 42	5,094
CUB val	remaining 15%; used for all model selection	900
CUB test	official test split, evaluated once after the recipe was frozen	5,794
Long-tail train	exponential subsample of CUB train, 26 $\rightarrow$ 3 images per class	2,039
iNat geo train / val	156 species, outside western North America; outings kept together	32,083 / 5,664
iNat geo test	western North America, at least 100 km from any training photo	18,953
CUB bg train / val	CUB images in 4 background clusters (foliage, perches, sky, water)	8,015 / 1,423
CUB bg test	held-out “ground / grass” background cluster, 196 species	2,350

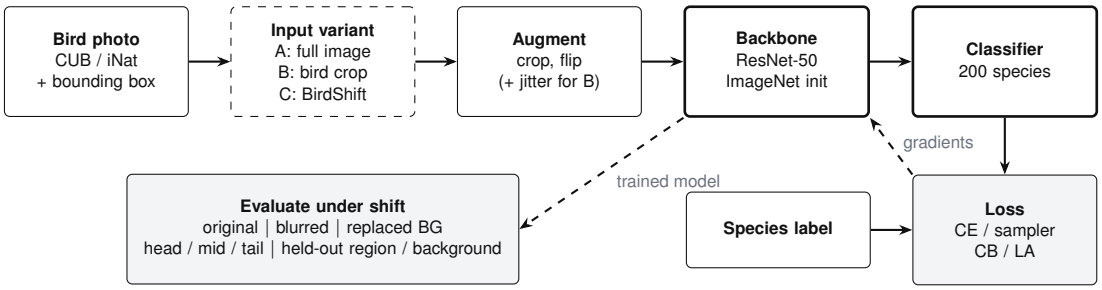
Kept fixed

- ResNet-50 backbone, 200-class head, seed 42
- Checkpoint = best val; held-out splits: SGD recipe, 448 px

Metrics

- Top-1, macro-F1, balanced (mean per-species) accuracy
- Accuracy change under each shift; head / medium / tail accuracy

Architecture and Pipeline

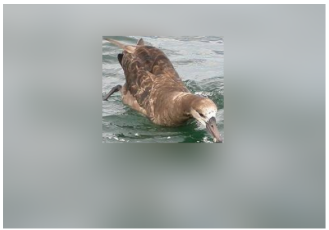


BirdShift (variant C)	Imbalance methods	Speed-ups
Outside the box is blurred (radius 22) with $p = 0.5$ per training image. No box is needed at test time.	Weighted sampler, class-balanced CE (β=0.999), logit-adjusted CE (τ=1), post-hoc LA.	float16 mixed precision + `channels_last`: 448 px run in about 12 min on one RTX 3090.

Background shift

Experiment 1: same bird, only the context changes







Original
Normal evaluation. Reference accuracy.

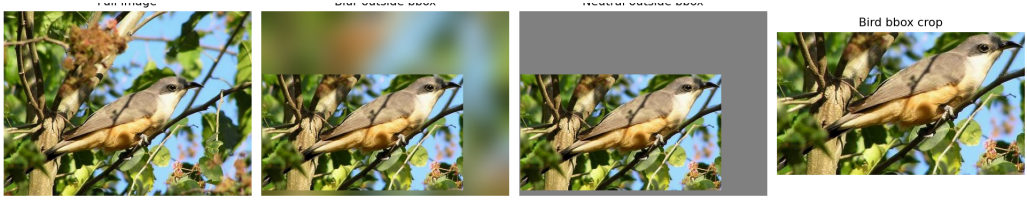
Blurred background
Gaussian blur (radius 22) outside the box. Colours stay, texture goes.

Replaced background
Flat neutral gray. No donor photo, so no second bird or habitat can leak in.

If accuracy drops when only the background changes, the model was partly using the scene.

BirdShift training

Break the link between background and label

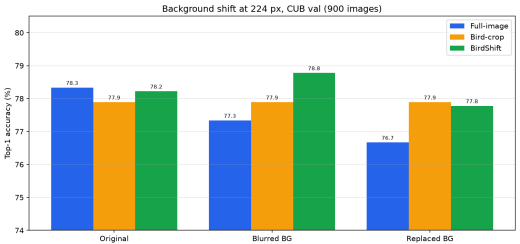


- Each training image: with probability 0.5 the area outside the bounding box is blurred.
- The model sees the same bird with and without its habitat, so the habitat stops being a reliable cue.
- At test time the input is the plain full image, like variant A.
- Bird crop (variant B) needs a box for every test image; BirdShift does not.

Results at 224 px

Experiment 1: top-1 on the 900 validation images

Model	Original	Blurred	Replaced
Full image	**78.33**	77.33	76.67
Bird crop	77.89	77.89	77.89
BirdShift	78.22	**78.78**	**77.78**
Change from original (pp)			
Full image		−1.00	−1.67
Bird crop		0.00	0.00
BirdShift		+0.56	−0.44

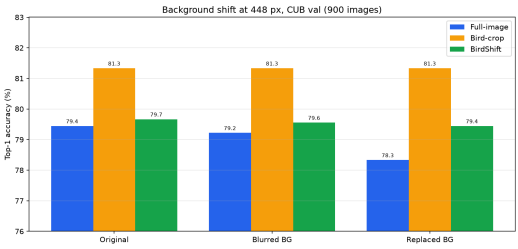


- Full image loses accuracy whenever the background changes.
- BirdShift matches it on normal photos (−0.11 pp) and holds up under both shifts.

Results at 448 px

Experiment 1: higher resolution, same conclusion

Model	Orig.	Blur	Repl.	vs 224
Full image	79.44	79.22	78.33	+1.11
Bird crop	**81.33**	**81.33**	**81.33**	+3.44
BirdShift	79.67	79.56	79.44	+1.45
Change from original (pp)				
Full image		−0.22	−1.11	
BirdShift		−0.11	−0.23	



- Every model gains from 448 px; the crop model gains most.
- Full image still drops most under replacement (10 images); BirdShift moves by 2.
- BirdShift beats full image in all three conditions.

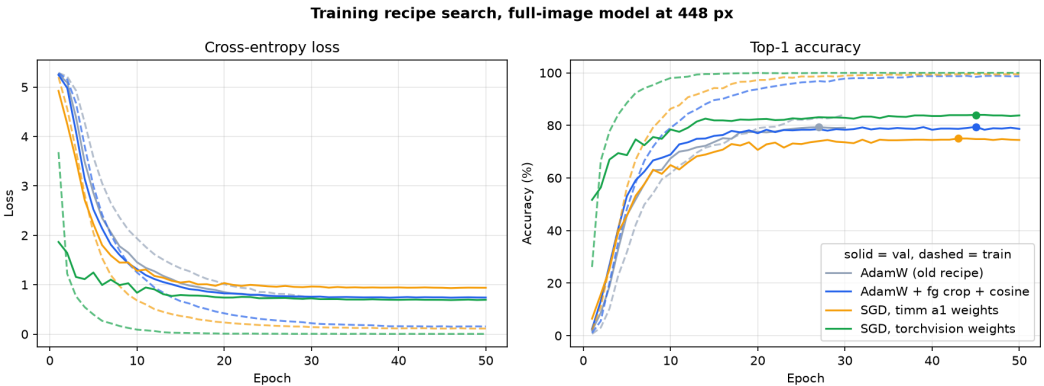
Reaching the published baseline

ResNet-50 at 448 px, one change at a time

Run	Optimizer	Weights	Crop	Schedule	Val	Test
First recipe	AdamW 10^{-4}	timm a1	RandomResizedCrop	constant, 30 ep	79.33	80.38
+ crop, cosine	AdamW 10^{-4}	timm a1	Resize 512 + crop 448	cosine, 50 ep	79.44	80.74
+ SGD	SGD 0.01, head $\times 10$	timm a1	Resize 512 + crop 448	cosine, 50 ep	75.00	76.08
+ torchvision weights	SGD 0.01, head $\times 10$	torchvision	Resize 512 + crop 448	cosine, 50 ep	**84.00**	**83.88**
Final: train + val	same recipe, all 5,994 training images, last epoch				n/a	**84.55**

- **Pretrained weights were the main factor** (+7.8 pp). timm's default weights come from a heavy-augmentation, BCE-loss recipe and transfer worse to birds [5, 6].
- SGD only helps together with the torchvision weights.
- Recipe chosen on val; test evaluated once at the end.
- Published reference: 84.5% [3]. Model released as `v1.0` on GitHub; this recipe is used for Experiments 3 and 4.

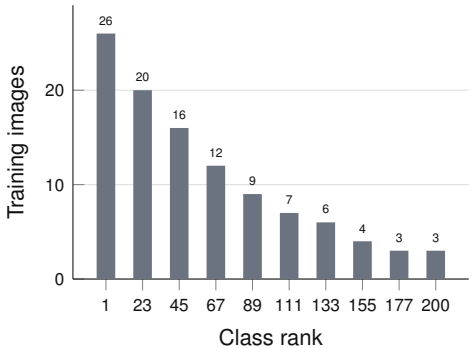
Recipe search: training curves



Solid = validation, dashed = train, dot = best epoch. The torchvision-initialised run starts at 52% val after one epoch and stays about 5 pp above the others throughout.

Class imbalance

Experiment 2: CUB is balanced, so we make it long-tailed



$n_r = n_{\max} \cdot 10^{-r/199}$: 5,094 → 2,039 images. Classes shuffled first, so the tail is not one bird family. Val stays balanced.

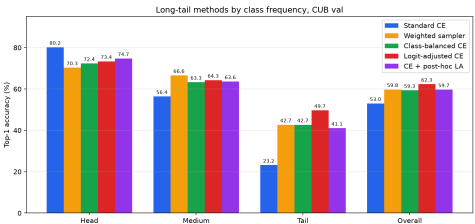
Method	**Idea**
Standard CE	plain cross-entropy
Weighted sampler	sample image $i \propto 1/n_{c(i)}$
Class-balanced CE	$w_c = \frac{1-\beta}{1-\beta^{n_c}}$, $\beta = 0.999$ [2]
Logit-adjusted CE	train on $z + \tau \log \pi$, predict with z [4]
Post-hoc LA	CE model, predict with $z - \tau \log \pi$

π_c = share of training images in class c; $\tau = 1$ as in the paper, not tuned. Head / medium / tail = thirds of classes by training count.

Long-tail results

Experiment 2: top-1 on the balanced validation set

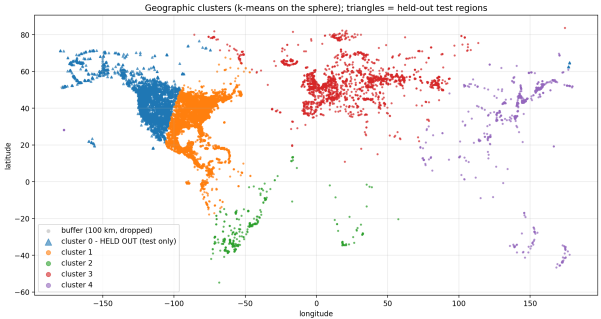
Method	Top-1	F1	Head	Mid	Tail
Standard CE	53.00	.477	**80.20**	56.39	23.18
Weighted sampler	59.78	.571	70.31	**66.56**	42.72
Class-bal. CE	59.33	.563	72.35	63.28	42.72
Logit-adj. CE	**62.33**	**.597**	73.38	64.26	**49.67**
Post-hoc LA	59.67	.567	74.74	63.61	41.06



- Plain CE: 80% on common classes, 23% on rare ones.
- **Logit-adjusted CE is best**: +9.3 pp overall, tail accuracy more than doubled.
- Post-hoc LA needs no retraining and still gives +6.7 pp.

Geography split

Experiment 3: iNaturalist photos of CUB species (Ankit)



- Research-grade, CC-licensed, location within 10 km.
- k-means (k=5) on lat/lon unit vectors.
- Test = western North America; 100 km buffer removed from train/val.
- 156 species, 56,700 photos; whole outings kept on one side.

Fixed in review

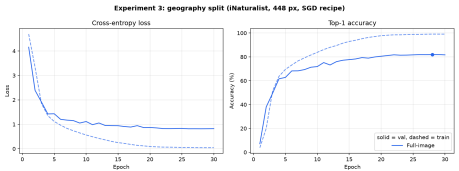
- 7 flock photos with 2–3 species labels dropped; duplicates merged.
- Myrtle Warbler restricted (no Audubon's); ambiguous *Sayornis* excluded.

Held-out region results

Experiment 3: test on unseen western North America

Model	**Val** (5,664)	**Test** (18,953)	**Drop**
CUB model, not retrained	49.91	46.93	−3.0
Trained on geo split	**81.94**	**62.68**	−19.3
balanced accuracy	77.84	73.08	−4.8

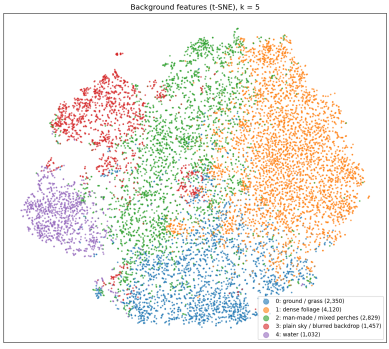
- CUB → iNaturalist photos costs ∼35 pp: photo style matters more than region.
- **Most of the drop is species mix.** Val reweighted to test species counts: 70.25%. So 11.7 pp label shift, 7.6 pp the region itself.
- Common test species are rare in training (2–7 val images): Rhinoceros Auklet 11.6%, Pelagic Cormorant 26.6%, Heermann Gull 37.1%.

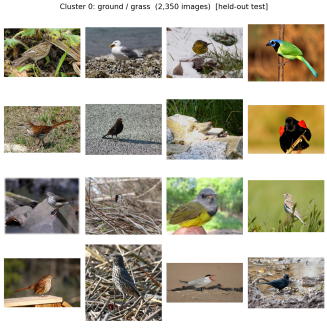


Full-image model only: iNaturalist has no bounding boxes, so crop and BirdShift cannot be trained here.

Background split

Experiment 4: cluster CUB images by background (Agastya)





Held-out cluster: ground / grass

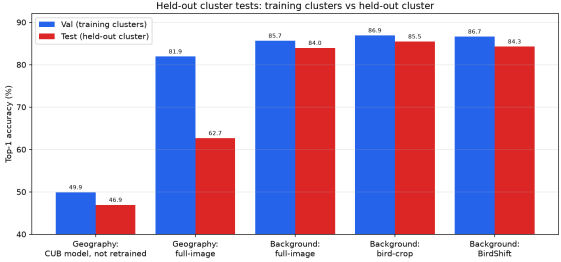
- Bird box masked, ResNet-50 features of the background only.
- k-means, k=5: foliage, perches, sky, water, ground / grass.
- Test = ground / grass (2,350); train / val from the rest.
- Ignores CUB's official split, so all models retrained.

Held-out background results

Experiment 4: all three models, SGD recipe at 448 px

Model	**Val** (1,423)	**Test** (2,350)	**Drop**
Full image	85.66	83.96	−1.7
Bird crop	**86.93**	**85.49**	−1.4
BirdShift	86.65	84.30	−2.4

- An unseen background costs every model <2.5 pp, far less than an unseen region (-19.3).
- Bird crop best on test (+1.5 pp, 36 images) but needs a box.
- BirdShift +0.3 pp (8 images) over full image: within one-seed noise.



Problems and fixes

Problem	How we noticed	Fix
Bird-crop model stuck at 40%	Far below full image (78%)	Pad crops to square before resizing, add augmentation: 40.3 → **77.9%**
Two evaluators could disagree	Shift test must equal normal eval	Image-by-image audit: 0 mismatches
Replaced background could leak	Donor photos bring habitat or birds	Use flat neutral gray instead
BirdShift overwrote crop checkpoints	Wrong mode stored in a checkpoint	Removed override, one folder per run
CUB too balanced for imbalance study	25–30 images per class	Seeded 10:1 long-tail subsample
4 pp below published ResNet-50	80.4% vs 84.5% on test	Ablation: torchvision weights + SGD → **84.55%**
Same photo, several species (iNat)	Duplicate `photo_ids` in review	Drop multi-label photos, assert uniqueness
Background clusters renumbered on rerun	Rerun on another GPU permuted ids	Assign to saved centroids: 99.1% agreement

Current progress

Done

- 3 input variants, 224 and 448 px
- Synthetic background-shift test
- Long-tail split, 5 methods
- Recipe matching 84.5% on CUB test
- Held-out region test (56.7 k iNat)
- Held-out background test, 3 models
- Code, results, model on GitHub

Findings

- Synthetic shift: BirdShift most stable without needing a box
- Real unseen background: small drop for all models
- Unseen region: large drop, mostly species mix

Next

- Rerun Experiments 1 and 2 with the SGD recipe
- Label-shift correction for the geography test (prior reweighting)
- Logit adjustment + BirdShift together
- Test-split numbers, 3 seeds, paired significance tests

All numbers: one seed (42). Experiments 1 and 2 on the 900-image val split (1 pp = 9 images); Experiment 4 test: 1 pp = 24 images.

References

[1] C. Wah et al. *The Caltech-UCSD Birds-200-2011 Dataset.* Caltech, 2011.

[2] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, S. Belongie. *Class-Balanced Loss Based on Effective Number of Samples.* CVPR 2019.

[3] Z. Yang et al. *Learning to Navigate for Fine-grained Classification.* ECCV 2018.

[4] A. K. Menon et al. *Long-Tail Learning via Logit Adjustment.* ICLR 2021.

[5] R. Wightman, H. Touvron, H. Jégou. *ResNet Strikes Back: An Improved Training Procedure in timm.* NeurIPS 2021 Workshop.

[6] S. Kornblith, J. Shlens, Q. V. Le. *Do Better ImageNet Models Transfer Better?* CVPR 2019.

[7] K. Xiao, L. Engstrom, A. Ilyas, A. Madry. *Noise or Signal: The Role of Image Backgrounds in Object Recognition.* ICLR 2021.

[8] K. He, X. Zhang, S. Ren, J. Sun. *Deep Residual Learning for Image Recognition.* CVPR 2016.

[9] P. Micikevicius et al. *Mixed Precision Training.* ICLR 2018.

[10] I. Loshchilov, F. Hutter. *SGDR: Stochastic Gradient Descent with Warm Restarts.* ICLR 2017.

[11] G. Van Horn et al. *The iNaturalist Species Classification and Detection Dataset.* CVPR 2018.

Code, results and the trained model: `github.com/Shafwansafi06/BirdShift`